

KISS principle

Most Linux distros and programs follow the KISS principle – Keep It Short and Simple

Scope Creep

Scope Creep is the steadily increasing scope of a single software. Adobe Photoshop is a good example of this

Feature

multi-threaded downloading. These features ensure that there is no need to install additional programs for downloading data. A download manager should ideally be able to download flash videos from a number of web sites. If the download manager can handle torrents, even better, the need for a torrent client is removed. Free Download Manager has these features packed in, but retains a small size despite this. The application is very utilitarian, and users will discover that they eventually end up using most of the features Free Download Manager has to offer. We would be interested if a download manager could additionally connect to P2P servers, as well as capture streaming audio. This is one of those cases where additional features actually reduces by feature bloat, by limiting the number of software needed for the task at hand.

Internet software

Microsoft Outlook is slow to start up, heavy on the system resources, and has nervous ticks if it is alive for long durations of time. A great alternative is Eudora, which manages most of the functions of Outlook quite well. Mozilla Thunderbird is another great application, but suffers from having a few superfluous features. These however, include a calendar and a manager, which is something that quite a few find useful. Eudora is the choice for strictly mail related matters, Thunderbird is the way to go for a personal information manager (PIM) along with an email client.

Instead of Gtalk, MSN Messenger and Yahoo! Messenger, a great alternative IM client is Pidgin. Pidgin is easy to use, has a simple interface, and is portable, which gives it an advantage over the other free multi-network chat client, Miranda.

Document handling

Although MS Office is pretty bloated, the alternatives such as Open Office or Lotus Symphony are simply not as smooth, and easy to handle as MS Office. Moreover, Open Office suffers from feature bloat simply because it has to handle a whole range of competing formats, and does not handle all of these formats well enough. A simple to use, and low footprint word processor is Abiword. This is a no-nonsense word processor, built for a single purpose, and is not part of an office suite. Google Docs, with their spreadsheet and word program, are great alternatives to an installed office suite. The collaboration aspect greatly improves the appeal of Google Docs.

Adobe Acrobat Reader is notorious for being bloatware in the past few releases. While all it was expected to do was open a

particular kind of document, the file format itself started accommodating different kinds of data, which the reader grew to handle. Adobe Acrobat Reader is probably the best example there is, of a software that has been in active development for too long. Even the commentary on the documents carried back and forth can have embedded videos. Commentary itself is an obscure enough feature for most users, video commentary is going too far. Users can also web-conference while reading a document. A simpler and understandably, much smaller PDF reader is Foxit reader.

Editing

Typically, users tend to have an image editor, an image batch processor, and a software for remote hosting. This is apart from the software that came with their cameras. This is accumulated bloat, and it is best to un-install all of them, then start over with the programs and features that you most require.

Picasa 3 should ideally solve all three needs. It has a basic range of image editing options available, handles batch resizing before uploading to the web, and is an image viewer with decent enough eye-candy. If adding a watermark, or bat-processing images in many ways is necessary, then IrfanView is a very small software, that nonetheless has a surprising number of features built in. IrfanView also allows for plug-ins for added functionality. For advanced image editing, both Gimp and Paint.Net are great alternatives

to Photoshop. Gimp is a little difficult to figure out, and needs a lot of plug-ins before the functionality reaches anywhere close to Photoshop, but has a few tricks up its sleeve that even Photoshop does not. Paint.Net is very easy to figure out, has a familiar user interface, and the plug-ins are easy to install. Our suggestion for the amateur: Paint.Net.

Both Audacity and Wavepad are great when it comes to sound editing, but the functionality that

Wavepad offers in comparison to the size of the download is unmatched. A 50 kb download, and you get to record, mix, and add a host of effects to your track.

A bloat-free future

The original vision for all software was a highly modular approach. The basic software would do very little, it was upto the user to find or code in the required

Bullet-point Engineering

The process of looking at competitor's feature list and incorporating them in the developer's own feature list is known as Bullet-point Engineering. An unfortunate tendency of opinion leaders (read tech journalism), involved tables of competing software, with points given to the most features. The size of the software, or the amount of memory it consumed, were not given enough weight as the number of features in the software. Naturally, the coders for the software tried to fit in as many features as possible into the software, to gain an edge over their competitors.

plugins and add-ons. Computers however, were used in far more ways than was thought possible when they were first introduced to the general public. In a

mad scramble to appease the dumb consumption of software by a large mass of people who couldn't bother to train themselves to use computers intelligently, issues similar to software bloat appeared. Remember that users are actually paying for the bloated software. The cost of coding authentication algorithms, developing all the features you one does not use, and of the increasingly tough copy-protection, is all transferred to the person who purchases the software. Reducing bloat is something that both the developer's and the users want, but finding

a commercially viable alternative is the problem.

A simple way to handle bloat in the future is to use remotely hosted applications, similar to Google apps. Such a setup would reduce bloat in a lot of ways. Consider a set-up like this: the end user has nothing but a screen and a high-bandwidth connection to a remote server. The server has all the hardware necessary to run the software, and a single instance of the software itself. Even if most users use only a few of the features, since the software is centrally located, there is little or no wastage. Of course, the interface for the software will be highly customised, so that the individual users don't waste the space on their screen with features they don't use. Next, as part of the development cycle, the software is updated in one place only, standardising the formats across all the users.

Till such systems come into place, regular users have little choice in choosing the features as well as the software they want to use. The problem with the assumption that users have a steadily increasing amount of hard disk space and memory to use, is that the users are forced to keep upgrading their system. Perfectly sound hardware for most purposes gets obsolete quickly for this reason. A computer purchased in the mid eighties lasted five years without any hardware upgrades. The same cannot be said for a computer purchased in the mid nineties. The kind of modularity envisioned for all operating systems, is true to an extent for the Linux OS, which is a great option for reviving and putting to good use an "obsolete" system. 

